Cybercrime and Ethical Hacking

Module I

Introduction of cybercrime

Cybercrime refers to any criminal activity that involves a computer, network, or networked device.

Term encompasses a broad range of illegal behaviors, including fraud, identity theft, and the distribution of illegal materials, all executed through digital means.

The impact of cybercrime can be devastating, leading to financial losses, data breaches, and compromised personal information.

Classifications of Cybercrime

Email Spoofing: Email spoofing is the forgery of an email header to make the message appear to have been sent by someone other than the actual source.

Spamming: Spamming refers to the practice of sending unsolicited bulk emails, often for commercial purposes. It can be used to spread malware, conduct phishing attacks, or simply to annoy recipients.

Cyber Defamation: Cyber defamation involves the use of electronic media to damage someone's reputation by making false or derogatory statements. This can include posting libelous content on websites, social media platforms, or in emails

Internet Time Theft: Internet time theft occurs when an individual uses someone else's internet connection without their knowledge or consent, often to conduct illegal activities online.

Salami Attack/Salami Technique: The salami attack or salami technique involves making a series of small, seemingly insignificant changes to data, often for financial gain. For example, a hacker might steal small amounts of money from many different accounts to avoid detection.

Data Diddling: Data diddling refers to the unauthorized alteration of data before or during input into a computer system, often for the purpose of fraud or sabotage

Forgery: Cybercrime forgery involves the creation of fake digital documents, such as certificates or contracts, to deceive others or gain unauthorized access to systems.

Newsgroup and Usenet Crimes: Crimes emanating from Usenet newsgroups include the posting of illegal or offensive content, harassment, and the distribution of copyrighted material without permission

Industrial Spying/Industrial Espionage: Industrial spying or espionage involves the theft of trade secrets, proprietary information, or intellectual property from a competitor or business partner using electronic means.

Hacking: Hacking refers to the unauthorized access to computer systems or networks, often with the intent to steal data, disrupt operations, or gain control of the system.

Computer Sabotage: Computer sabotage involves the intentional disruption or destruction of computer systems or networks, often through the use of malware or denial-of-service attacks.

Email Bombing/Mail Bombs: Email bombing or mail bombs refer to the practice of sending a large number of emails to a single address, often with the intent to overwhelm the recipient's inbox or cause system crashes.

Computer Network Intrusions: Computer network intrusions involve the unauthorized access to private or government computer networks, often with the intent to steal sensitive data or disrupt operations

Password Sniffing: Password sniffing is the act of intercepting and capturing passwords transmitted over a network, often using packet sniffers or other monitoring tools.

Credit Card Frauds: Credit card frauds involve the unauthorized use of credit card information to make purchases or obtain cash, often through the use of skimming devices or phishing scams

Identity Theft: Identity theft occurs when an individual's personal information, such as their name, social security number, or financial information, is stolen and used for fraudulent purposes, such as opening new accounts or making unauthorized transactions.

Cyberstalking: Harassment that utilizes electronic means to stalk or intimidate an individual. It can manifest in various ways and is characterized by malicious intent, often causing significant distress to victims

Software Piracy: Software piracy is the illegal use or copy of paid software with violation of copyrights or license restrictions.

Online Drug Trafficking: With the big rise of cryptocurrency technology, it became easy to transfer money in a secured private way and complete drug deals without drawing the attention of law enforcement. This led to a rise in drug marketing on the internet.

Cyber Extortion: Cyber extortion is the demand for money by cybercriminals to give back some important data they've stolen or stop doing malicious activities such as denial of service attacks.

Types of Cybercriminal

Hackers: Individuals who exploit vulnerabilities in systems. They can be categorized as:

Black Hat Hackers: Malicious hackers who exploit systems for personal gain. - **White Hat Hackers:** Ethical hackers who test systems for vulnerabilities with permission.

Grey Hat Hackers: Operate between ethical and unethical hacking, sometimes exploiting vulnerabilities without malicious intent but without permission.

Fraudsters: Individuals who commit financial crimes online, such as credit card fraud or investment scams.

Cyber Terrorists: Individuals or groups that use cyber attacks to instill fear or cause disruption, often motivated by political or ideological goals.

Organized Cybercrime Groups: These groups operate like traditional criminal organizations, often providing "crime as a service" and engaging in high-level cyber operations for profit.

Role of computer in Cybercrime

Computers serve a major role in crime which is usually referred to as "Cybercrime".

This cybercrime is performed by a knowledgeable computer user who is usually referred to as a "hacker", who illegally browses or steals a company's information or a piece of individual private information and uses this information for malevolent uses.

In some cases, this person or group of individuals may become evil and they destroy and corrupt data files.

This cyber or computer-based crime is also known as hi-tech crime or electronic crime.

As the computer is the main source of communication across the world, thus this can be used as a source of stealing information and this information can be used for their benefit.

Prevention of Cybercrime

1. Be sure that you are using up-to-date security software like antivirus and firewalls.

2. Implement the best possible security settings and implementations for your environment.
3. Don't browse untrusted websites and be careful when downloading unknown files, and also be careful when viewing Email attachments.
4. Use strong authentication methods and keep your passwords as strong as possible.
5. Don’t share sensitive information online or on your social media accounts.
6. Educate your children about the risks of internet usage and keep monitoring their activities.
7. Always be ready to make an immediate reaction when falling victim to cybercrime by referring to the police.

Ethical Hacking

- Ethical hacking is the authorized practice of testing systems, networks, and applications to identify and fix security weaknesses before malicious attackers exploit them.

- Ethical hackers use real hacking techniques with permission to strengthen cybersecurity and protect digital assets

Goals of Ethical Hacking

- Legal and professional security testing
- Ethical hackers simulate real-world cyberattacks
- Helps identify vulnerabilities and weak points
- Strengthens an organization’s overall security posture
- Supports compliance with cybersecurity standards
- Builds resilience against modern cyber threats

Phases of Ethical Hacking

- **1. Reconnaissance (Investigation)**
- Gathering information on the target system (e.g., IP addresses, operating systems, open ports).
- **Tools**: Nmap, Whois, Google Dorks.
- **2. Scanning**
- Vulnerability identification in the target system (e.g., open ports, services, misconfigurations).
- **Tools**: Nessus, OpenVAS, Nikto.
- **3. Gaining Access**
- illegal access to the system using vulnerabilities.
- **Techniques**: Password cracking, exploiting software vulnerabilities, social engineering.
- **4. Maintaining Access**
- Guaranteed ongoing access to the system (e.g., installing backdoors, rootkits).
- **Tools**: Metasploit, Netcat.
- **5. Clearing Tracks**
- Erasure of attack traces to avoid detection.
- **Techniques**: Log deletion, timestamp manipulation.

Hacker and Cracker & Phreaker

Feature	Hacker (Ethical)	Cracker (Malicious)	Phreaker
Goal	Protect, improve, learn	Steal, damage, profit	Exploit phone systems
Ethics	Ethical (White Hat)	Unethical (Black Hat)	Varies, often exploitative
Focus	Computers, Networks	Computers, Networks	Telephone Systems
Legality	Often Legal	Illegal	Illegal (Toll Fraud)

Rules of Ethical Hacking

Scope and Objectives

Authorization and Permission

Confidentiality, Integrity, and Availability (CIA Triad)

Vulnerability Assessment

Technical Proficiency

Analytical and Problem-Solving Skills

Continuous Learning

Balancing Security and Privacy

Keeping Up with Evolving Threats